ARTIFICIAL INTELLIGENCE-BASED CHATBOT For Appliance Control

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This project presents an artificial intelligence (AI)-based chatbot for controlling electrical home appliances using ESP8266-12E NodeMCU and Facebook Messenger platform. The platform for creating AI chatbot for Facebook is Chatfuel. The project also uses IFTTT and Adafruit IO for back-end support. The project is great for integrating AI with the Internet of Things (IoT).

The block diagram shown in Fig. 1 explains the process and setup required for this project. The IoT and chatbot are developed separately, and then merged to make the final chatbot.

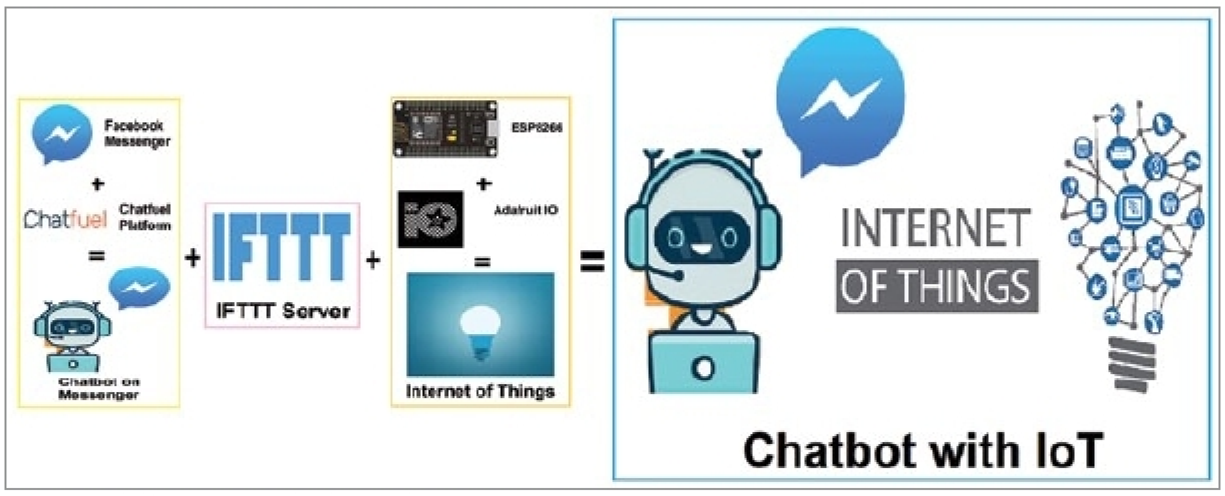

Fig. 1: Block diagram of AI-based chatbot

Circuit diagram

The circuit diagram for the AI-based chatbot is shown in Fig. 2. ESP8266-12E NodeMCU manufactured by Lolin is used in the project. Check the datasheet if you are using a different version.

USB Type-C cable is used to program NodeMCU from a laptop or PC. The relay module is an electromechanical switching device. You can control AC or DC appliances digitally by providing control input to the relay pins. Here, a 5V, single-channel, one-changeover relay is used. Connections to an AC appliance (AC bulb) through the relay module are shown in the circuit diagram.

Adafruit IO is linked to NodeMCU at the hardware side. Output of Chatfuel is integrated with IFTTT and Adafruit IO servers. Output pins of NodeMCU are controlled through Facebook Messenger chatbot. Signals from NodeMCU control the relay, which, in turn, controls the electrical appliance, such as AC lamp or bulb.

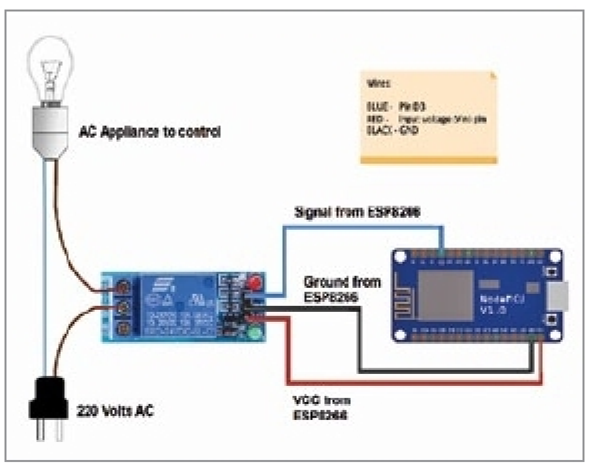

Fig. 2: Circuit diagram of AI-based chatbot

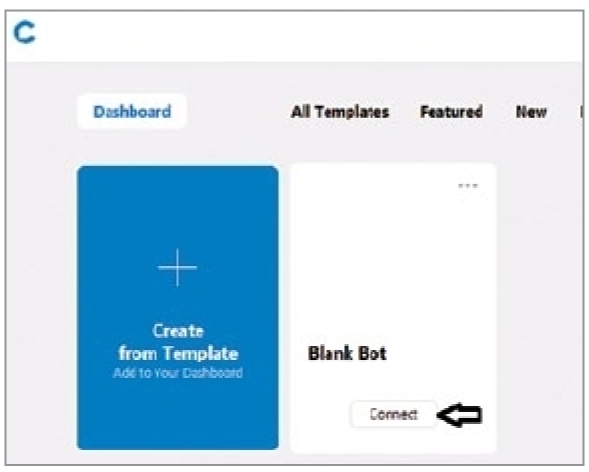

Fig. 4: Create a chatbot

Facebook Messenger setup

A virtual human is needed to build the chatbot. For this, create a Facebook

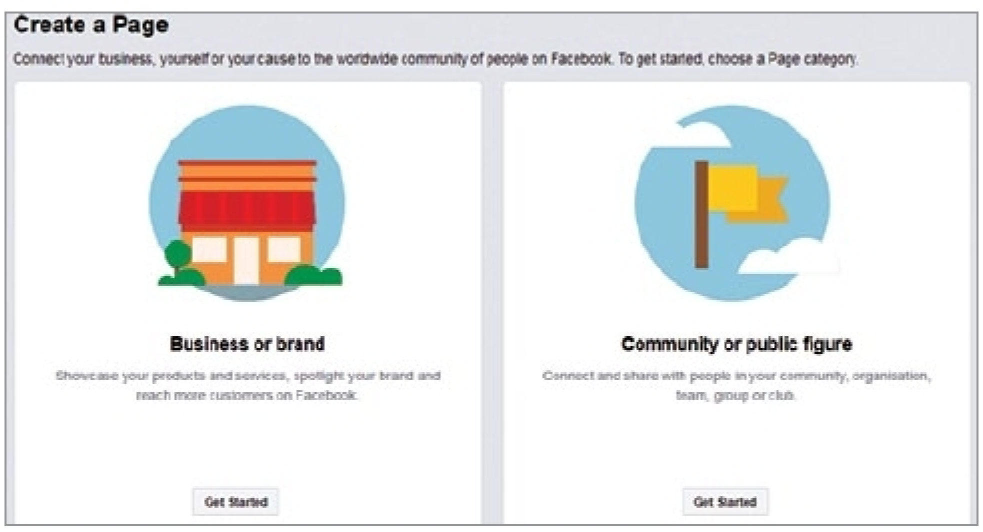

Fig. 3: Create Facebook page (community or public figure)

Fig. 5: Edit Welcome Message